

Mock interview

1. NLP is a branch of AI that helps computers understand human language.
2. The main goal of NLP is to understand, process, and generate human language.
3. NLP is used in chatbots, translation, sentiment analysis and speech recognition.
4. NLP is the overall field, NLU understands language and NLG generates language.
5. Major challenges include ambiguity, context, language variations.
6. Human language variations is difficult because it is complex and context-dependent.
7. Ambiguity means a word or sentence can have multiple meanings.
8. Lexical is word ambiguity, syntactic is structure ambiguity, semantic is meaning ambiguity.
9. A language model predicts words or sequence of words.
10. Rule-based NLP uses rules, while statistical NLP learns from data.
11. A regular expression is a pattern used to match text.
12. Common symbols are *, +, ?, [], ^ and \$.
13. * means zero or more occurrences.
14. + means one or more occurrences.
15. ^ allows zero occurrences, while + requires at least one.

16. Character classes match characters from a specific set.
17. An FSA is a machine that recognizes specific patterns using state.
18. FSA components are states, transitions, input, start state, and final state.
19. FSA is used for tokenization and pattern matching in NLP.
20. DFA has one transition for an input, while NFA can have multiple transitions.
21. Morphology is the study of word structure.
22. A morpheme is the smallest meaningful unit of language.
23. A free morpheme stands alone; a bound morpheme attaches to another word.
24. Inflectional morphology changes the grammatical form of a word.
25. Derivational morphology creates a new word.
26. Inflection changes word form, derivation creates a new word.
27. Ex: Play $\rightarrow$ Played.
28. Ex: Happy $\rightarrow$ Happiness.
29. Morphological parsing breaks words into meaningful parts.
30. Finite-state parsing is used to analyze word structures efficiently.
31. Stemming reduces words to their root form.
32. Lemmatization converts words to their meaningful base form.

33. Stemming cuts words, while lemmatization gives the proper base word.
34. Porter Stemmer is a popular algorithm used for English stemming.
35. It may sometimes produce incorrect or meaningless stems.
36. An N-gram is a sequence of N words or tokens.
37. A unigram contains one word.
38. A bigram contains two consecutive words.
39. A trigram contains three consecutive words.
40. Unigram has 1 word, bigram has 2, and trigram has 3 words.
41. $P(\text{word}_2 | \text{word}_1) = \text{count}(\text{word}_1, \text{word}_2) / \text{count}(\text{word}_1)$.
42. N-grams help predict words based on previous words.
43. An unsmoothed N-gram uses only observed data.
44. Unseen N-grams receive zero probability.
45. Smoothing avoids zero probabilities.
46. Smoothing assigns probability to unseen events.
47. Backoff uses lower-order N-grams when higher-order N-grams are missing.
48. Deleted interpolation combines unigram, bigram and trigram probabilities.
49. Entropy measures uncertainty in predicting language.
50. High entropy indicates high uncertainty.
51. POS tagging assigns grammatical tags to words.
52. POS tagging helps understand sentence structure.

53. Major word classes include noun, pronoun, verb, adjective and adverb.
54. A noun names a person, place, thing or idea.
55. A pronoun is used instead of a noun.
56. A verb shows an action or state.
57. An adjective describes a noun.
58. An adverb describes a verb, adjective or another adverb.
59. A determiner identifies a noun, like a on the.
60. A preposition shows relationships, like in, on or under.
61. A conjunction joins words or sentences.
62. A POS tagset is a collection of grammatical tags.
63. POS ambiguity occurs when a word has multiple grammatical roles.
64. Rule-based POS tagging uses grammar rules.
65. Stochastic POS tagging uses probability and statistics.
66. Transformation based POS tagging learns rules to correct tags.
67. Rule-based uses fixed rules, while stochastic uses probabilities.
68. The same word can have different tags depending on context.
69. "Read a book" = noun, "Book a ticket" = verb.
70. Context helps identify the correct POS tag.
71. A CFG is a grammar used to describe sentence structure.

72. CFG contains terminals, non-terminals, rules and a start symbol.
73. A terminal symbol is a final word or token.
74. A non-terminal symbol represents a grammatical category.
75. A Production rule defines sentence structure.
76. A Parse tree shows the grammatical structure of a sentence.
77. Both show sentence structure, a syntax tree focuses on syntax.
78. $S \rightarrow NP VP$ means a sentence consists of a noun phrase and verb phrase.
79. Syntactic analysis studies the grammatical structure of sentences.
80. Parsing analyzes a sentence using grammar rules.
81. Top-down parsing starts from the start symbol.
82. Bottom-up parsing starts from the input words.
83. Top-down starts with grammar, while bottom-up starts with words.
84. An Earley Parser is a dynamic parsing algorithm for CFGs.
85. It is useful for complex and ambiguous grammars.
86. Grammatical agreement means matching grammatical features between words.
87. Subject-verb agreement means the verb matches the subject in number.

88. Subcategorization shows what complements a word can take.
89. A feature structure contains linguistic feature & their values.
90. A PCFG is a CFG with Probabilities assigned to its rules.
91. Discourse is a sequence of connected sentences.
92. Discourse analysis studies language beyond individual sentences.
93. Reference resolution finds what a word refers to.
94. Anaphora resolution identifies what a Pronoun refers to.
Ex: "John came. He Sat."
95. Text coherence means logical connection between ideas.
96. Discourse Structure is the organization of ideas in a text.
97. Cohesion links sentences; coherence creates overall meaning.
98. A dialogue system communicates with users using language.
99. A conversational agent is a chatbot that interacts with users.
100. A dialogue act is the Purpose of an utterance, such as a question or request.